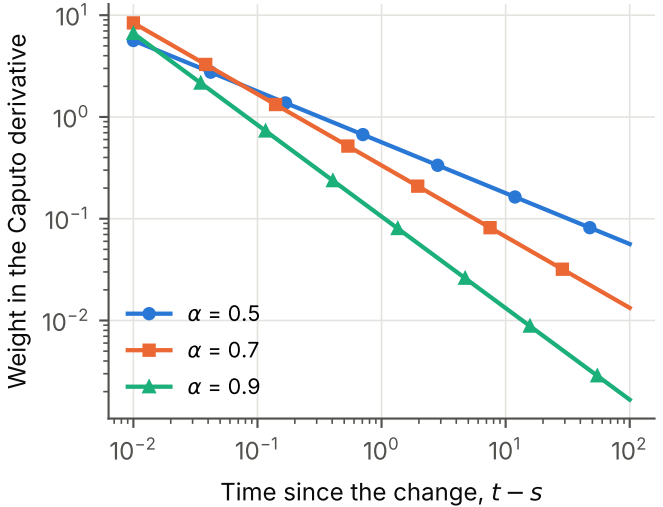
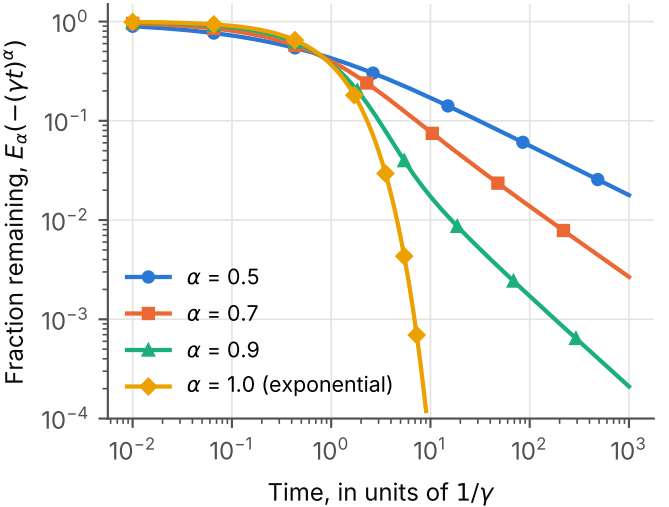

(a) memory: weight of past changes
Weight in the Caputo derivative
10^{1}
10^{0}
10^{-1}
10^{-2}
α = 0.5
α = 0.7
α = 0.9
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Time since the change, $t - s$


(b) slow, power-law decay
Fraction remaining, $E_\alpha(-(\gamma t)^\alpha)$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
α = 0.5
α = 0.7
α = 0.9
α = 1.0 (exponential)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Time, in units of $1/\gamma$